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Early-stage lung cancer in the era of lung cancer screening: VA outperforms other insurance models

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Abstract

Background: Lung cancer screening guidelines were introduced in the United States in 2013, with variable implementation. This study aimed to evaluate temporal diagnostic trends in non-small cell lung cancer (NSCLC) diagnosis since the introduction of these guidelines.

Methods: This retrospective cohort analysis used the VA Corporate Data Warehouse and the National Cancer Database. We evaluated temporal trends in the distribution of NSCLC stage at the time of diagnosis, including differences based on insurance coverage type (including uninsured, privately insured, Medicare, Medicaid, and VA coverage) with adjustment for clinically relevant variables.

Results: Among 1,450,965 patients diagnosed from 2006–2020, the proportion of NSCLC cases diagnosed at Stage I increased in all insurance groups (by 12.74%, 2%, 0.25%, and 2.57% for the VA, Medicare, Private Insurance, and Medicaid, respectively). If all insurance systems achieved the unadjusted stage distribution seen in the Veterans Health Administration, an additional 45,684 patients would be diagnosed with Stage I NSCLC and 65,933 fewer patients would be diagnosed with stage IV disease.

Conclusions: For patients with any form of insurance, there has been an increase in the proportion of early-stage NSCLC (Stage I and II) and a corresponding decrease in the proportion of Stage III and IV since the introduction of national lung cancer screening guidelines. As the largest integrated “single-payer” healthcare system in the US, the VA dramatically outperforms

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other insurance types, perhaps attributable to universal coverage and robust lung cancer screening programs.

INTRODUCTION:

Lung cancer remains one of the most common cancer types and accounts for the most cancer-related deaths in the United States and globally, with incidence and mortality trends determined largely by prior tobacco exposure.[1–3] Although lung cancer mortality has declined in the U.S.[4], survival rates remain low[1], with known disparities based on race, sex, and socioeconomic status.[5]

Clinical outcomes for non-small cell lung cancer (NSCLC) are directly related to stage at the time of diagnosis, with drastically superior survival for early stages (Stage I and II) compared to late stages (Stage III and IV)[6,7]. Complete resection is the standard of care for early-stage NSCLC[8] and therefore, early detection offers the best opportunity for long-term survival and cure.

Two large randomized control trials, the U.S. National Lung Cancer Screening Trial (NLST)[9] and the European NELSON Trial[10], demonstrated the utility of early detection with reduced lung-cancer mortality in screened individuals. These results prompted the U.S. Preventive Services Task Force (USPSTF) to release the first U.S. national lung cancer screening guidelines in December 2013, with a grade B recommendation for annual low-dose chest CT scans for adults with heavy smoking histories[11]. Since then, essentially every expert consensus group has released similar guidelines aimed at early detection, but it has been noted that these guidelines may not capture all high-risk individuals.[12]

Between 2013 and 2016, the Veterans Health Administration (VHA) conducted the VA Lung Cancer Screening Demonstration Project to study the feasibility of implementing a comprehensive screening program in eight pilot sites; annual adherence to screening guidelines was 64%[13] and cancer detection rate was 1.5%, with 71% of cancers detected at stage 1 or 2.[14] Despite these early promising results and expanded coverage for the Veteran and civilian populations, uptake of guidelines has been poor across the U.S.[15], with significant disparities in access to screening and surveillance.[16,17] Cited implementation barriers include cognitive factors (lack of guideline awareness from providers and patients, limited information/misinformation, low perceived risk, limited patient acceptance due to cost), access factors (logistical issues, cost, financial/social impediments), and psychological factors (fear, fatalism, stigma).[18–20] To address cost concerns, the Centers for Medicare and Medicaid Services (CMS) expanded coverage to include lung cancer screening for the civilian population in February 2015.[21]

Our study aimed to evaluate temporal trends in early-stage NSCLC diagnosis relative to the introduction of national screening guidelines with comparison across different insurance types. We hypothesized that the proportion of NSCLC cases diagnosed at early stages would increase over time despite suboptimal implementation of screening guidelines and that these trends would vary based on insurance type.

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METHODS:

Study Population

This retrospective cohort study used civilian and Veteran patient data from the National Cancer Database (NCDB) 2021 participant user file (released 2023) and the VA Informatics and Computing Infrastructure (VINCI), which collates clinical and administrative data from the VA Corporate Data Warehouse, respectively[22]. The NCDB, a database supported by the American Cancer Society and the American College of Surgeons, captures greater than 70% of newly diagnosed cancers in the United States[23] and although it is not truly "population-based", it is highly representative of the U.S. population[24], with data for patients insured through all major civilian providers and uninsured patients.

NSCLC cases were identified using International Classification of Diseases (ICD) for Oncology (third edition) codes. Staging data were harvested from the NCDB and VINCI and categorized according to the American Joint Commission on Cancer Staging Manual (8th edition). We included all patients diagnosed with NSCLC from 2006–2020 and excluded those with insurance coverage through "other government insurance" or other/unknown histologic classification, clinical or pathologic stage, insurance coverage type, race, or socioeconomic status.

Covariates:

NCDB variables were defined according to the Facility Oncology Data Standards and the Standards for Oncology Registry Entry (v2018). Variables common to the VA were defined according to the same data standards as the NCDB[25] and extracted for further analysis. Demographic variables included age, sex, race, clinical or pathologic stage at the time of diagnosis, and residential zip code at the time of diagnosis. As previously described[25], zip codes were linked to obtain county-level median household income and education level (proportion of adults with at least a high school diploma), and comorbidities were determined using ICD 9/10 codes to calculate a composite Charlson/Deyo score.

Outcomes

The primary outcome was Stage 1 NSCLC diagnosis (as a proportion of all NSCLC cases) and the secondary outcome was overall distribution of NSCLC stages. These endpoints were compared between patients covered by different types of insurance (including No Insurance, Private Insurance, Medicare, Medicaid, and the VA).

Statistical Analysis

Data were analyzed using SAS version 9.4 (SAS Institute, Cary, NC) and baseline cohort descriptive statistics were reported. Demographic differences between groups were assessed using chi-square and T-test analysis with Bonferroni adjustment. P-values<0.05 were considered statistically significant.

Univariate analysis evaluated the proportion of NSCLC cases diagnosed at Stage I, with comparison between different insurance types. The unadjusted proportion of Stage I NSCLC

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was plotted against time and Cochrane-Armitage trend analysis was performed to determine statistical significance.

Adjusted multivariable logistic regression determined the annual adjusted proportion of NSCLC cases diagnosed at Stage I (with 95% confidence intervals) for a standardized 65-year-old white male patient with no comorbidities, low educational attainment and income, and urban home location. The restricted cubic spline method[26] was applied with the following knots: 2007, 2013, and 2019, corresponding to the 10th, 50th, and 90th percentiles, respectively.

Temporal and Geospatial Analysis

To assess stage migration over time, the annual distribution of all NSCLC stages was calculated (with each reported as an unadjusted proportion) and reported for each insurance type.

Geospatial analysis was performed to assess for trends within the VA system and the entire NCDB civilian cohort. The unadjusted proportion of Stage I cases was determined for each US Census division[27] and averaged over three time blocks within the study period relative to the 2013 USPSTF screening guidelines; 1) Before: 2006–2013, 2) Immediately After: 2014–2017, and 3) Late After: 2018–2020. Each region was compared to the national average using 2-tailed t-test. Data were plotted separately for the VA and NCDB cohorts using a single-color heat-map weighted for the proportion of Stage I NSCLC cases.

Idealized Simulation

After determination of the VA as the leader in Stage I diagnosis, we performed a simulation to examine the theoretical benefit of adopting the VA system for all patients. The annual unadjusted proportions of each stage observed in the VA was applied to all other insurance groups to determine the theoretical NSCLC stage distribution in the civilian NCDB cohort.

RESULTS:

After application of inclusion/exclusion criteria (Supplemental Figure 1), 1,450,965 patients with NSCLC were analyzed (Medicare: 863,018; Private Insurance: 345,848; Medicaid: 91,669; VA: 89,907; and No Insurance: 37,84). Characteristics for the unadjusted cohorts are presented in Table 1, with significant baseline differences in age, sex, race, socioeconomic status, comorbidity burden, and home rurality based on insurance type (all p-values <0.0001).

Over the study period (2006–2020), the total number of NSCLC cases decreased for all insurance types (Supplemental Table 1). Annual unadjusted proportion of Stage I NSCLC was plotted (Figure 1), demonstrating that prior to 2012, Medicare had the highest proportion of Stage I, followed sequentially by the VA, Private Insurance, Medicaid and No Insurance (Figure 1). Cochrane-Armitage trend analysis revealed that Stage I NSCLC increased proportionally over time for insured patients but not uninsured patients (p<0.00001 for all insurance groups, p=0.7029 for No Insurance). In 2013, the VA surpassed Medicare as the leader in Stage I NSCLC diagnosis.

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Adjusted analysis demonstrated similar temporal trends (Figure 2), with the VA achieving leadership in Stage I NSCLC diagnosis in 2009. From 2006–2020, Stage I NSCLC increased proportionally from 26.84% to 39.58% (VA), 29.23% to 31.23% (Medicare), 24.05 to 24.30% (Private Insurance), and 16.38% to 18.95% (Medicaid). In the No Insurance group, Stage I NSCLC decreased proportionally from 12.74% to 10.98% over the same period but was not statistically significant.

When examining the annual unadjusted distribution of all NSCLC stages, we note that with a temporal increase in Stage I and II diagnoses (early-stage), there was a concomitant decrease in Stage III and IV (late-stage). This trend was apparent in all insurance groups, but most pronounced in the VA cohort (Figure 3 and Supplemental Figure 2).

Geospatial analysis of regional trends in Stage I NSCLC diagnosis also demonstrated migration toward Stage I over time in all regions but overall, the VA had less geographic variability than the NCDB cohort (Supplemental Figure 3). Detailed comparative statistics are provided in Supplemental Table 2.

In a simulation designed to examine the theoretical benefit of implementing the VA system for all patients, we see that if all insurance systems achieved the unadjusted NSCLC stage distribution observed in the VA, 45,485 additional cases would be diagnosed at Stage I (as opposed to more advanced stages) and 65,933 fewer cases would be diagnosed at Stage IV over the study period (Supplemental Figure 4).

COMMENT:

In this study, we show that from 2006–2020, there has been migration toward earlier-stage NSCLC diagnosis for insured patients. This effect is most pronounced within the VA system, where we see a 12.74% increase in the proportion of Stage I NSCLC over the study period and a corresponding decrease in Stage IV disease. In other insurance groups, this migration has been more modest (0.25–2.57% over the same period). This represents a promising trend, as TNM stage at diagnosis is the single most important prognostic determinant for NSCLC, with an inverse correlation between stage at diagnosis and long-term survival. Patients diagnosed at Stage I and II are candidates for treatment with curative intent, with 70–90% 5-year survival.[28,29]

Given these temporal trends relative to the 2013 introduction of national screening guidelines, we posit that CT screening has contributed to NSCLC stage migration. However, our analysis does not examine the effect of these guidelines directly, as our cohort includes patients already diagnosed with lung cancer and not the screening-eligible population. We could not limit analysis to patients eligible for screening under the current guidelines based on the absence of smoking history in the NCDB, nor could we examine the modality by which patients were diagnosed. Over the last two decades, there has been greater use of cross-sectional imaging and undoubtedly, some patients were diagnosed through means other than low-dose CT screening. Regardless, we observe a temporal trend toward earlier diagnosis that corresponds to the introduction of national guidelines aimed at early lung cancer detection in high-risk patients.

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The American College of Chest Physicians and the American Thoracic Society have outlined elements of ideal screening programs including multidisciplinary specialists experienced in nodule management and the diagnosis and treatment of lung cancer.[30] Although our study is not designed to understand causality, the VA's comprehensive lung cancer screening program incorporates this multidisciplinary model and includes certain practices that may promote early lung cancer diagnosis through effective management of screen-detected results. The VA screening program includes appropriate patient selection and education resources within the primary care system, tobacco cessation resources, efficient communication of screening results, and a nurse navigator to assist with screening implementation. These services are housed within the VHA's Lung Precision Oncology Program[31], which has designated centralized hubs across the U.S. that coordinate care (screening, genetic testing, and clinical trial enrollment) for Veterans through a hub-and-spoke model and in 2020, the VHA initiated a public-private partnership with the GO_2 Foundation for Lung cancer designed to expand access to screening through additional screening sites, raise screening awareness, and share best practices[32]. Additionally, the VHA provides transportation services and meal vouchers to offset the burdens associated with accessing care. Care coordinators send appointment reminders and if a patient misses an appointment, they are rescheduled without penalty. Notably, these services are all provided free of charge to Veterans, thereby mitigating cost as a hindrance to screening and follow-up for detected abnormalities and likely contribute to reduced racial disparities in the VA's surgical treatment for resectable NSCLC.[33]

While we show the VA to be a leader in Stage I NSCLC diagnosis, prior studies have described barriers to accessing high-quality lung cancer care within the VA system, including social and economic disadvantages, fragmented care, and limited health literacy among Veterans.[34] Screening implementation and management of screen-detected results remains suboptimal in both the civilian and Veteran patient populations and should be the focus of future research and policy reform.

The U.S. may draw lessons from successful screening programs abroad. In South Korea, there is governmental buy-in for lung cancer screening and the centralized healthcare system covers at least 90% of screening costs for everyone and up to 100% for low-income patients.[35] Even under these idealized circumstances, only 23% of eligible patient are screened annually, compared to 5.8% of eligible patients in the U.S.[36] This highlights the difficulty of achieving meaningful adherence to screening guidelines. In the original NLST, adherence to annual follow-up was nearly 95% and in modeling studies that evaluated the potential benefit of USPSTF guidelines, the authors presumed 100% adherence.[37] A more pragmatic benchmark for lung cancer screening should reflect the reality of cancer screening in the U.S., as reflected in national breast and colorectal cancer screening programs.[38,39] Understanding the causes of suboptimal adherence and input from the field of dissemination and implementation science to develop multi-pronged solutions will likely incrementally improve screening outcomes[40] and contribute toward further advances in early-detection of lung cancer. Additionally, targeting smoking cessation while broadening screening access is essential to continued improvements in lung cancer outcomes.

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The current study has some limitations. First, retrospective analysis is subject to sampling bias. Second, analysis was limited by the available data; smoking status is not captured by the NCDB, and therefore we could not adjust for smoking status in our model. We have adjusted for the available variables that differ significantly between insurance groups, however there may be residual confounding that is not captured. Third, in both databases, staging data is variably documented leading us to categorize patients based on the available staging data (clinical or pathologic). Nevertheless, analysis included a large, nationally representative data set with a study period encompassing the introduction of screening guidelines and therefore provides valuable insights into trends in NSCLC diagnosis.

We showed that among patients with NSCLC, there has been temporal migration toward earlier-stage NSCLC diagnosis for patients with any form of insurance from 2006–2020, and the VA dramatically outperforms other insurance types regarding the proportion of patients diagnosed at Stage I. The biggest effect is observed after the 2013 introduction of lung cancer screening guidelines, despite known barriers to screening program implementation. However, our analysis includes only patients already diagnosed with lung cancer and therefore, we cannot comment directly on overall rates of screening implementation or whether the observed stage migration is attributable to effective screening. In future studies, we plan to evaluate features of lung cancer screening implementation within the VA which can be adopted by the civilian healthcare system.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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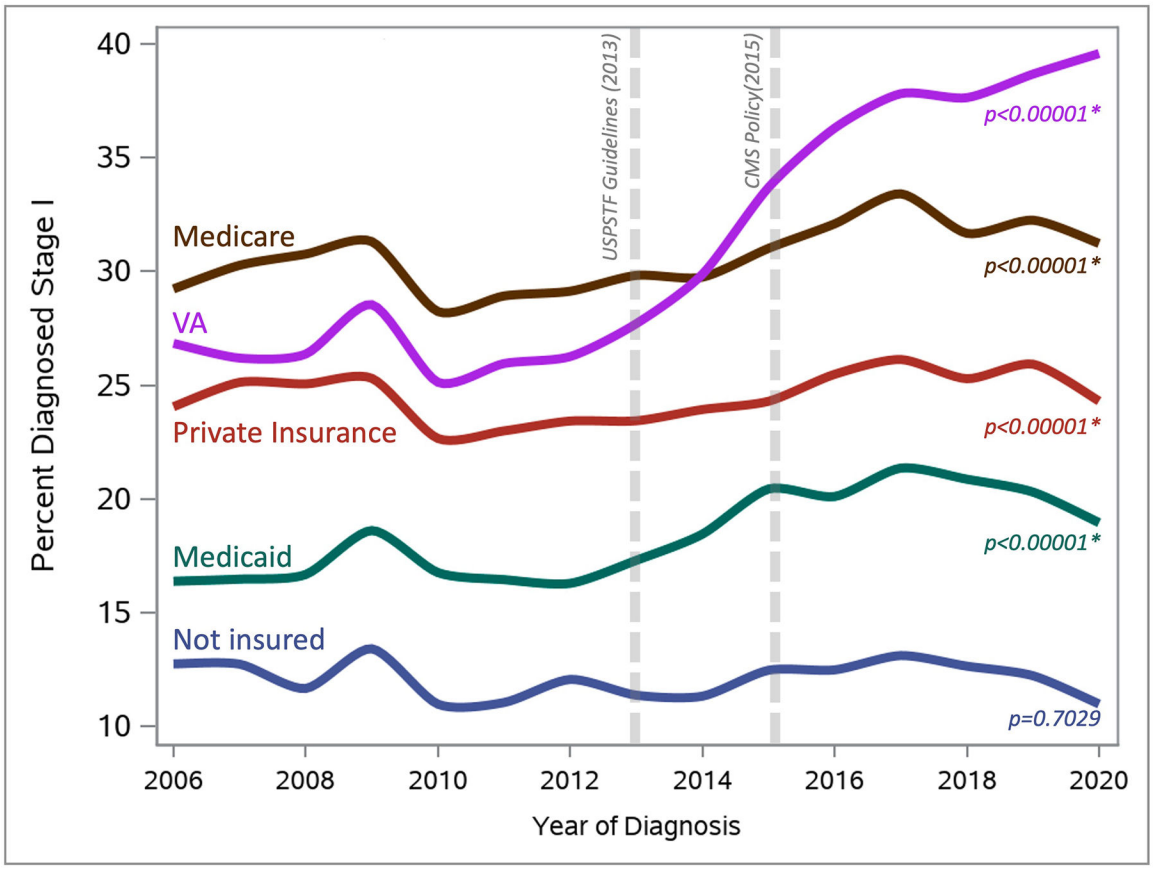


Figure 1 -

Annual unadjusted rate of Stage I NSCLC diagnosis (as a proportion of all cases), with p-values reported for Cochrane-Armitage trend analysis.

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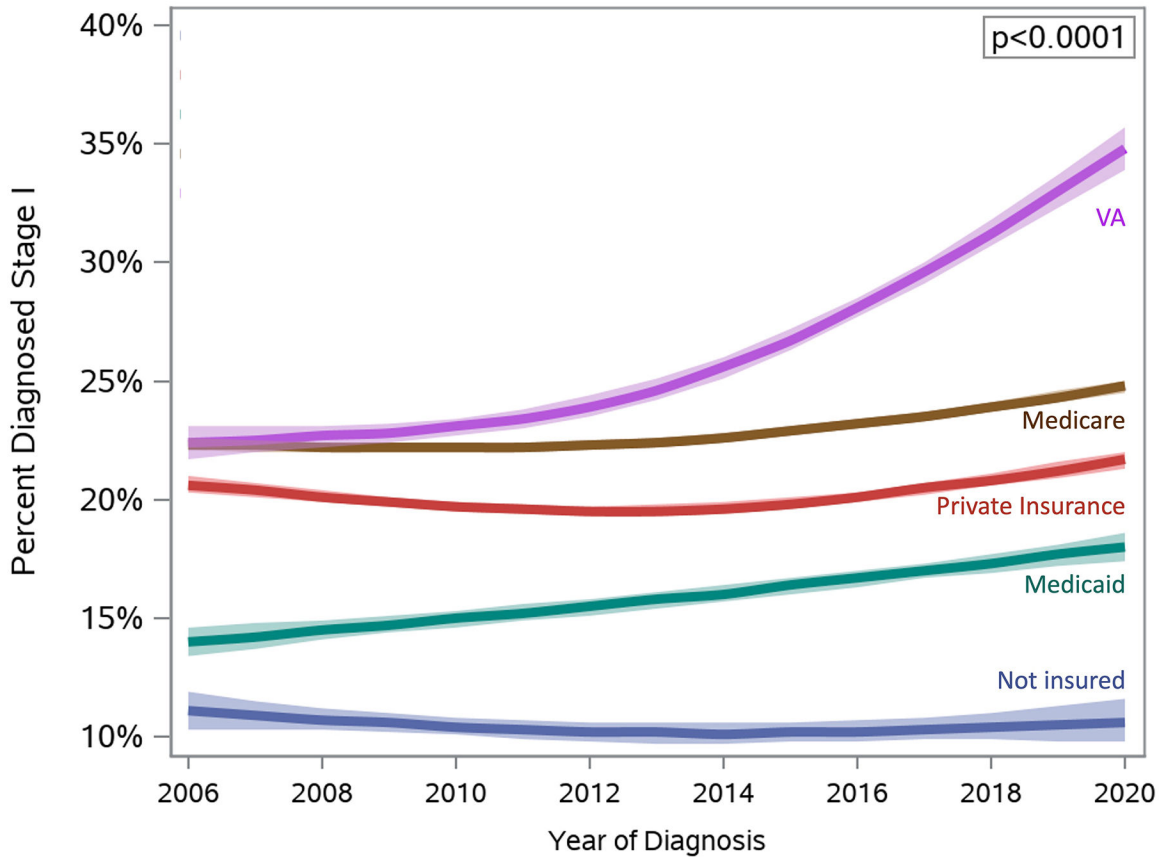


Figure 2 -

Annual adjusted rate of Stage I NSCLC (as a proportion of all cases) for a standardized 65-year-old white male patient with no comorbidities and a low education, low income, urban home location. The restricted cubic spline method was applied with the following knots: 2007, 2013, and 2019, corresponding to the 10th, 50th, and 90th percentiles, respectively. Trends are reported with 95% confidence intervals.

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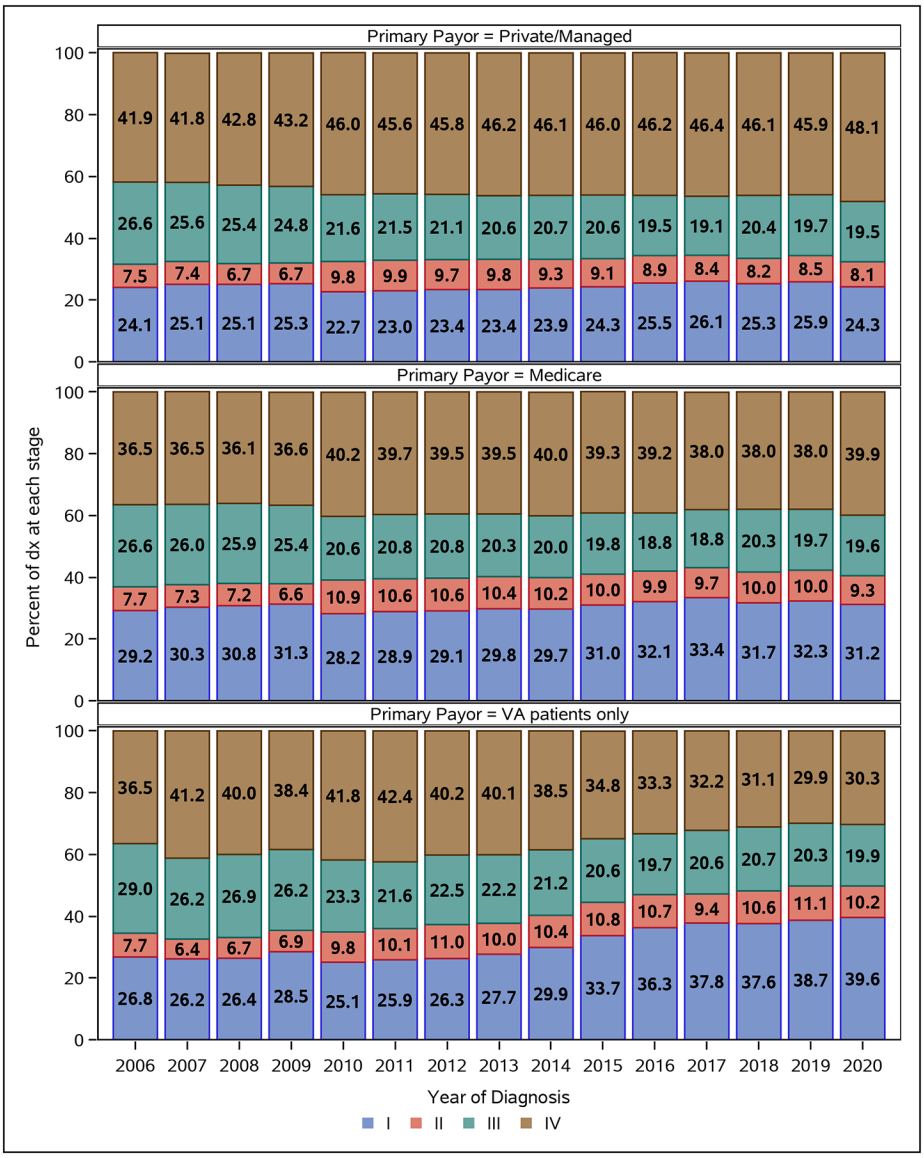


Figure 3 –.

Annual unadjusted distribution of all NSCLC stages (with each reported as a proportion of all cases) for patients with Private Insurance (top panel), Medicare (middle panel), and the VA (bottom panel).

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Table 1:

Demographics of the Study Population

Variable	VA (N=89,907)	Private (N=345,848)	Not Insured (N=37,846)	Medicare (N=863,018)	Medicaid (N=91,669)	*p-value*
Age, mean (SD), y	69.51 (8.90)	61.34 (9.60)	57.77 (8.94)	73.51 (7.92)	58.33 (9.06)	*<0.0001*
Sex, n (%)						
Female	2,361 (2.63)	169,257 (48.94)	15,707 (41.50)	415,454 (48.14)	42,449 (46.31)	*<0.0001*
Male	87,539 (97.37)	176,591 (51.06)	22,139 (58.50)	447,564 (51.86)	49,220 (53.69)	*<0.0001*
Race, n (%)						
Black	15,289 (17.01)	36,085 (10.43)	7,737 (20.44)	82,489 (9.56)	21,715 (23.69)	*<0.0001*
Other	1,095 (1.22)	14,865 (4.30)	1,861 (4.92)	24,059 (2.79)	6,202 (6.77)	*<0.0001*
White	73,523 (81.78)	294,898 (85.27)	28,248 (74.64)	756,470 (87.65)	63,752 (69.55)	*<0.0001*
SES group (median household income-educational attainment), n (%)						
Low-Low	39,889 (44.37)	101,259 (29.28)	18,945 (50.06)	286,431 (33.19)	45,914 (50.09)	*<0.0001*
Low-High	10,167 (11.31)	47,042 (13.60)	4,837 (12.78)	100,672 (11.67)	11,807 (12.88)	*<0.0001*
High-Low	11,386 (12.66)	32,723 (9.46)	3,481 (9.20)	96,373 (11.17)	9,571 (10.44)	*<0.0001*
High-High	28,465 (31.66)	168,824 (47.66)	10,583 (27.96)	379,542 (43.98)	24,377 (26.59)	*<0.0001*
Charleson/Deyo Comorbidity Score, n (%)						
0	28,375 (31.56)	225,464 (65.19)	24,423 (64.53)	461,916 (53.52)	53,266 (58.11)	*<0.0001*
1	23,389 (26.01)	83,777 (24.22)	9,620 (25.42)	240,206 (27.83)	24,644 (26.88)	*<0.0001*
2	15,594 (17.34)	25,502 (7.37)	2,580 (6.82)	101,855 (11.80)	8,555 (9.33)	*<0.0001*
3+	22,549 (25.08)	11,105 (3.21)	1,223 (3.23)	59,041 (6.84)	5,204 (5.68)	*<0.0001*
Home Location, n (%)						
Rural	2.647 (2.94)	5,028 (1.45)	865 (2.29)	18,419 (2.13)	1,937 (2.11)	*<0.0001*
Urban	86,653 (96.38)	330,380 (95.53)	36,271 (95.84)	822,503 (95.31)	88,391 (96.42)	*<0.0001*
Unknown	607 (0.68)	10,440 (3.02)	710 (1.88)	22,096 (2.56)	1,341 (1.46)	*<0.0001*
NSCLC Stage, n (%)						
I	27,689 (30.80)	84,640 (24.47)	4,557 (12.04)	264,579 (30.66)	17,042 (18.59)	*<0.0001*
II	8,483 (9.44)	29,463 (8.52)	2,265 (5.98)	81,288 (9.42)	6,796 (7.41)	*<0.0001*
III	20,408 (22.70)	75,812 (21.92)	8,590 (22.70)	184,725 (21.40)	21,040 (22.95)	*<0.0001*
IV	33,327 (37.07)	155,933 (45.09)	22,434 (59.28)	332,426 (38.52)	46,791 (51.04)	*<0.0001*

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